

An Explainable Deep Learning Framework for Forest and Non-Forest Classification of Aerial Imagery Using a Custom CNN and MobileNetV2 Transfer Learning for Deforestation Monitoring

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Abstract

Deforestation is one of the most pressing environmental challenges of the present era, contributing directly to biodiversity loss, carbon emissions, and the degradation of ecological balance. Manual interpretation of aerial and satellite imagery for forest monitoring is slow, costly, and difficult to scale across large geographic regions. Although deep learning has been applied to remote-sensing imagery, many studies rely on a single architecture, evaluate models using accuracy alone, and provide little interpretability of the visual evidence driving model decisions. This study addresses these gaps by developing and comparing an explainable convolutional neural network (CNN) framework for binary Forest versus Non-Forest classification of aerial imagery. A custom CNN baseline is trained from scratch and benchmarked against a MobileNetV2 transfer-learning model on a publicly available dataset of 5,108 aerial images. Both models are evaluated using accuracy, precision, recall, F1-score, the area under the receiver operating characteristic curve, and confusion matrices, while Grad-CAM and SHAP are used to inspect the spatial regions that influence predictions. The custom CNN achieved a test accuracy of 82.1% with a macro-averaged F1-score of 0.79 and an AUC of 0.883, outperforming the MobileNetV2 model, which achieved 71.8% accuracy, a macro-averaged F1-score of 0.71, and an AUC of 0.818. The explainability analysis, combining Grad-CAM with SHAP, indicated that the better-performing model attended to genuine vegetation regions rather than background artefacts. These results show that a carefully designed custom CNN can outperform a general-purpose pre-trained network on a domain-specific

binary task while remaining computationally efficient and interpretable. This work contributes a reproducible, explainable, and resource-efficient pipeline for automated deforestation monitoring from aerial imagery.

Keywords: Deforestation monitoring, Convolutional neural networks, Transfer learning, MobileNetV2, Explainable AI, Grad-CAM, SHAP, Aerial image classification

1. Introduction

1.1. Application Background and Practical Importance

Forests cover roughly one third of the Earth’s land surface and provide ecological services that are essential to the stability of the global climate system. They regulate the carbon cycle, preserve biodiversity, protect watersheds, and sustain the livelihoods of communities that depend on forest resources. Deforestation, driven by logging, agricultural expansion, mining, and urbanisation, removes these services and accelerates climate change through the release of stored carbon [1]. The organisations affected by deforestation include environmental agencies, conservation bodies, policy makers, and local populations who rely on early and accurate information to respond to forest loss. Conventional monitoring depends heavily on manual interpretation of aerial and satellite imagery by trained analysts, which is labour-intensive, slow, and difficult to apply consistently over very large areas [2]. Delays or inaccuracies in identifying deforested regions allow illegal activity to continue unchecked and reduce the effectiveness of conservation interventions. Automated image analysis offers a scalable alternative that can process large volumes of imagery rapidly and flag areas of concern for human review. Recent advances in deep learning have made it possible to classify land cover from imagery with a level of accuracy that was previously unattainable using manual or rule-based methods [3]. For these reasons, an automated system capable of distinguishing forested from non-forested regions in aerial imagery has direct and practical value for environmental monitoring.

1.2. Machine Learning and Computer Vision Context

Image-based machine learning enables computers to extract meaningful information directly from visual data without relying on manually engineered features. Convolutional neural networks are particularly well suited to this

task because they learn hierarchical representations of images through successive layers of convolution and pooling [4]. Early layers capture low-level patterns such as edges, colours, and textures, while deeper layers combine these into higher-level structures that correspond to meaningful objects or regions. For forest classification, this capability is important because the visual distinction between forested and non-forested land lies in subtle combinations of colour, texture, and spatial structure. Unlike classical machine-learning approaches that require an expert to design features explicitly, CNNs discover the relevant features automatically during training [5]. This property reduces the dependence on domain-specific feature engineering and allows the same architecture to be adapted to many different image domains. However, deep networks typically require large quantities of labelled data to train effectively, and labelled remote-sensing imagery is often limited and expensive to produce. The challenge of limited labelled data is therefore a central consideration in the design of any practical forest classification system. This study addresses that challenge through careful preprocessing, data augmentation, and the use of transfer learning as a comparative baseline.

1.3. Transfer Learning for the Selected Problem

Training a deep convolutional network from random initialisation requires substantial data and computational resources, which are not always available for specialised domains such as forest monitoring. Transfer learning addresses this limitation by reusing networks that have already been trained on very large general-purpose image collections [6]. The features learned by such networks, including edges, textures, and shapes, are generic enough to be useful across many visual tasks, and they can be transferred to a new problem with relatively little additional training. A pre-trained network can be used either as a fixed feature extractor or as a starting point for fine-tuning on the target dataset. MobileNetV2 is a widely used architecture for this purpose because its depthwise-separable convolutions and inverted-residual structure provide strong accuracy at a low computational cost across diverse image-classification tasks [7]. Previous studies in land-cover and agricultural classification have reported that lightweight pre-trained networks such as MobileNetV2 can serve as efficient and effective baselines when adapted through transfer learning [7, 6]. This study uses MobileNetV2 as a transfer-learning comparison against a custom CNN in order to determine whether a general-purpose pre-trained network offers an advantage on this specific

binary task. A detailed technical description of each architecture is provided in the Methodology rather than here.

1.4. Explainable AI and Trustworthiness

A model that achieves high accuracy is not automatically trustworthy, because accuracy alone does not reveal whether the model is reasoning from meaningful evidence. A network may achieve good scores by exploiting background cues or dataset artefacts rather than the genuine visual characteristics of forest cover, a phenomenon known as shortcut learning. For environmental monitoring, where decisions can influence conservation policy and resource allocation, it is important to confirm that predictions are based on relevant image regions [8]. Grad-CAM addresses this need by producing class-specific heatmaps that indicate which spatial regions most strongly influenced a prediction [9]. SHAP complements this by attributing the contribution of individual image regions to the final output, providing a more granular view of model behaviour [10]. Together these methods allow the spatial reasoning of the model to be inspected and compared against domain expectations. If a model attends to vegetation regions when predicting the forest class, confidence in its decisions is justified, whereas attention to irrelevant regions signals a reliability problem. This study therefore integrates both Grad-CAM and SHAP to evaluate not only how accurately the models classify imagery but also whether their reasoning is visually sound.

1.5. Research Gap

A review of recent literature reveals several recurring limitations in deep-learning studies of forest and land-cover classification. Many studies evaluate only a single architecture and do not compare a custom network against a pre-trained alternative on the same data [2]. A large proportion of work reports accuracy as the primary or only metric, which can be misleading when classes are imbalanced [11]. Explainability is frequently absent, so it is often unclear whether reported performance reflects genuine visual understanding or incidental correlations [8]. Class imbalance, which is common in environmental datasets, is often not addressed explicitly. Few studies combine a comparative model evaluation with both Grad-CAM and SHAP interpretation in a single reproducible pipeline. This study addresses these gaps by comparing a custom CNN with a MobileNetV2 transfer-learning model, reporting a full set of evaluation metrics, and integrating two complementary explainability methods.

1.6. Problem Statement

This study addresses the problem of automatically classifying aerial images into forested and non-forested categories using a comparative and explainable CNN-based framework. The input consists of red-green-blue aerial images, and the prediction target is a binary label indicating the presence or absence of forest cover. The main modelling challenge is to learn a reliable decision boundary from a moderately sized and imbalanced dataset while avoiding reliance on background artefacts. The practical purpose of the system is to support scalable deforestation monitoring by flagging non-forested regions for further analysis. The study investigates whether transfer learning improves predictive performance over a custom CNN while maintaining computational efficiency and producing visually meaningful explanations.

1.7. Aim and Objectives

The aim of this study is to develop and evaluate an explainable CNN-based framework for the reliable classification of forested and non-forested regions from aerial imagery. The specific objectives are as follows:

- to analyse and preprocess a publicly available aerial-imagery dataset for binary forest classification;
- to develop a custom convolutional neural network as a baseline classifier;
- to implement a MobileNetV2 transfer-learning model for comparison;
- to compare predictive performance using accuracy, precision, recall, F1-score, AUC, and confusion matrices;
- to evaluate the training behaviour and computational efficiency of both models;
- to interpret model predictions using Grad-CAM and SHAP;
- to analyse errors, limitations, and the practical suitability of the framework for deployment.

1.8. Research Questions

- **RQ1:** Can a CNN-based framework reliably distinguish forested from non-forested regions in aerial imagery using only red-green-blue input?
- **RQ2:** Does MobileNetV2 transfer learning outperform a custom CNN trained from scratch on this binary forest-classification task?
- **RQ3:** Which evaluation metrics most accurately reflect model performance given the class imbalance present in the dataset?
- **RQ4:** Do Grad-CAM and SHAP confirm that the models base their predictions on genuine vegetation regions rather than background artefacts?
- **RQ5:** Which model offers the best balance between predictive accuracy, computational efficiency, and interpretability for practical deforestation monitoring?

1.9. Contributions and Novelty

This study makes several explicit contributions to applied deep learning for deforestation monitoring. First, it provides a reproducible end-to-end pipeline that covers data preparation, model development, evaluation, and explainability for binary forest classification. Second, it offers a systematic comparison between a custom CNN and a MobileNetV2 transfer-learning model on the same dataset under identical evaluation conditions. Third, it integrates two complementary explainability methods, Grad-CAM and SHAP, to verify the spatial reasoning of the models rather than relying on accuracy alone. Fourth, it demonstrates that a custom CNN can outperform a general-purpose pre-trained network on a domain-specific binary task, with implications for efficient deployment. The novelty of the work lies in the combination of comparative evaluation, full metric reporting, and dual explainability within a single resource-efficient framework, rather than in the use of any individual standard component.

1.10. Report Organization

The remainder of this report is organised as follows. Section 2 reviews related work on image-based forest monitoring, CNN methods, transfer learning, evaluation, and explainability. Section 3 describes the dataset, preprocessing, model architectures, training procedure, and evaluation framework.

Section 4 details the experimental setup and implementation environment. Section 5 presents the quantitative and explainability results. Section 6 interprets the findings and discusses limitations, and Section 7 concludes the report and outlines future work.

2. Literature Review

2.1. Image-Based Machine Learning in the Application Domain

The use of image analysis for environmental monitoring has evolved considerably over the past two decades. Early approaches relied on conventional image-processing techniques such as thresholding, spectral indices, and hand-crafted texture descriptors to distinguish land-cover types. These methods were later supplemented by classical machine-learning classifiers including support vector machines and random forests, which improved accuracy but still depended on manually designed features [3]. The transition toward deep learning marked a significant shift, as convolutional networks removed the need for explicit feature engineering and learned representations directly from data [2]. This shift enabled substantial gains in accuracy across a range of remote-sensing tasks, including land-cover mapping, change detection, and deforestation analysis. Studies have demonstrated that deep models capture subtle spatial and spectral patterns that are difficult to encode manually [12]. Nevertheless, challenges remain, particularly the dependence on large labelled datasets and the limited interpretability of complex models. These continuing challenges motivate the comparative and explainable approach adopted in the present study.

2.2. CNN-Based Methods for the Selected Task

Convolutional neural networks have become the dominant approach for forest and land-cover classification from imagery. De Bem et al. applied CNNs to Landsat data for change detection of deforestation in the Brazilian Amazon and reported strong classification performance [12]. Systematic reviews of deforestation detection confirm that CNN-based segmentation and classification consistently outperform classical methods across diverse forest ecosystems [2]. Studies in related domains such as forest-fire detection have similarly shown that CNN architectures including VGGNet, DenseNet, and MobileNet achieve high accuracy on aerial and satellite imagery [13]. Continuous monitoring of forest change has likewise benefited from satellite time-series analysis, which complements single-image classification [14]. Reported

dataset sizes vary widely, from a few thousand to tens of thousands of images, and class structures range from binary to multi-class land-cover taxonomies. Common preprocessing strategies include resizing, normalisation, and augmentation to improve generalisation [15]. A recurring weakness across these studies is the limited comparison between custom and pre-trained architectures, and the frequent reliance on accuracy as the sole evaluation metric. The present study responds to these methodological issues directly.

2.3. Transfer Learning Architectures

Transfer learning has been widely adopted in remote sensing to mitigate the scarcity of labelled data. Naushad et al. compared VGG16 and Wide Residual Networks for land-use classification on the EuroSAT dataset and demonstrated that fine-tuned pre-trained networks achieve high accuracy with reduced training effort [6]. MobileNetV2 is characterised by depthwise-separable convolutions and inverted residual blocks, which make it a highly parameter-efficient feature extractor well suited to resource-constrained settings. Other architectures such as VGG16 and ResNet variants have also been applied successfully to agricultural and land-cover tasks, though they typically carry substantially larger parameter counts than MobileNetV2 [6, 16]. Reviews of lightweight CNNs note that model size and inference cost are important considerations for deployment on resource-constrained platforms [4]. In this study, MobileNetV2 is selected as the transfer-learning comparison because of its established use as an efficient baseline and its compatibility with Grad-CAM and SHAP explainability. Only architectures directly relevant to the study are described in detail.

2.4. Data Preprocessing and Augmentation

Effective preprocessing is essential for stable and generalisable training of image classifiers. Common operations include resizing images to a fixed input dimension, normalising pixel values, and converting colour modes to a consistent representation [15]. Data augmentation, which generates transformed copies of training images, is widely used to expand effective dataset size and reduce overfitting [17]. Typical augmentations for aerial imagery include rotation, horizontal and vertical flipping, zoom, and brightness adjustment, all of which are valid because forest cover has no fixed orientation when viewed from above. Care must be taken to avoid transformations that destroy class-relevant information, although for overhead aerial imagery most geometric

transformations preserve the meaning of the scene. Class-balancing strategies such as class weighting, oversampling, and targeted augmentation are frequently used to address imbalance in environmental datasets. The present study applies resizing, normalisation, and a set of orientation-preserving augmentations, and addresses imbalance through class weighting during training.

2.5. Model Evaluation and Reliability

The choice of evaluation metric strongly influences the conclusions drawn from a classification study. Accuracy is intuitive but can be misleading when classes are imbalanced, because a model can achieve a high score by favouring the majority class [11]. Precision, recall, and the F1-score provide a more balanced view by accounting for false positives and false negatives, while the area under the receiver operating characteristic curve summarises performance across decision thresholds. The confusion matrix offers a detailed breakdown of correct and incorrect predictions for each class. Recent empirical work argues that a combination of metrics, rather than accuracy alone, is necessary for a trustworthy evaluation of predictive models [11]. Several studies in the remote-sensing literature continue to rely heavily on accuracy, which limits the reliability of their reported conclusions. This study reports a full set of complementary metrics to provide a robust assessment of both models.

2.6. Explainable AI in Image Analysis

2.6.1. Grad-CAM

Gradient-weighted Class Activation Mapping produces class-specific heatmaps by using the gradients flowing into the final convolutional layer to weight its activation maps. The resulting visualisation highlights the spatial regions that most influenced a particular prediction [9]. Grad-CAM has been used widely in medical imaging, agriculture, and remote sensing to confirm that models attend to meaningful regions [18]. Its main limitation is a relatively coarse spatial resolution, because the heatmaps are derived from the low-resolution feature maps of deep layers.

2.6.2. SHAP

SHapley Additive exPlanations attribute a contribution value to each input feature or region, indicating how strongly it supports or opposes a prediction. For images, SHAP is typically applied using masking or superpixel strategies to estimate the contribution of different regions [10]. The method

provides both positive and negative attributions, offering a more granular view than activation maps alone. Its principal drawback is computational cost, as accurate attribution requires many model evaluations.

2.6.3. Explainability Limitations

Visually appealing heatmaps are not proof of correct reasoning, and explainability outputs must be interpreted with caution. Attribution methods can be sensitive to preprocessing choices, may be unstable across runs, and depend on the specific model being explained [8]. They can occasionally highlight regions that are not semantically meaningful, which is why domain knowledge is required to interpret them. This study uses Grad-CAM and SHAP together so that the two methods can corroborate each other, reducing the risk of drawing conclusions from a single unreliable explanation.

2.7. Robustness, Generalization, and Deployment

Practical deployment of image classifiers requires attention to robustness and computational feasibility. Robustness to image noise, blur, illumination changes, and domain shift is an important consideration for real-world aerial imagery, although a full robustness study is beyond the present scope. Model size and inference time are important for deployment on edge devices and unmanned aerial platforms, where computational resources are limited [4]. Lightweight architectures are often preferred in such settings because they reduce memory and energy requirements while maintaining acceptable accuracy [7]. Although a full robustness study is beyond the scope of this work, computational efficiency is considered explicitly through the comparison of a compact custom CNN against the MobileNetV2 transfer-learning network.

2.8. Comparative Literature Review Matrix

Table 1 compares twelve closely related studies across datasets, model choices, evaluation practices, explainability, robustness, and validation strategies. The comparison shows that transfer learning is used frequently, but the combined use of multiple-model comparison, both Grad-CAM and SHAP explainability, and full metric reporting is rare. Most reviewed studies rely on accuracy and a limited set of metrics, and very few include SHAP-based attribution or external validation. The final row of the table represents the proposed study and shows how it addresses several of these recurring gaps simultaneously.

Table 1: Comparative literature review matrix summarising datasets, models, evaluation practices, explainability methods, robustness analysis, and validation strategies in closely related image-classification studies, with the proposed study shown in the final row. TL: Transfer Learning; CB: Class Balancing; MC: Multiple-Model Comparison; GC: Grad-CAM; SH: SHAP; RT: Robustness Testing; EV: External Validation; EA: Efficiency Analysis.

Study	Problem	Dataset	Cls	Size	Models	TL	CB	MC	GC	SH	RT	EV	EA	Metrics	Main Contribution	Principal Limitation
De Bem et al., 2020	Amazon deforestation change detection	Landsat Amazon	2	Large scene set	SharpMask, U-Net, ResUNet	✓	✗	✓	✗	✗	✗	~	✗	Acc, Kappa	✗ F1, ✗	No explainability or efficiency analysis
Jelas et al., 2024	Deforestation segmentation review	Multiple	Var	Survey	U-Net, SegNet, DeepLab	✓	~	✓	~	✗	~	✗	✗	IoU, Acc, F1	Systematic review of segmentation methods	No primary experiments
Khan et al., 2023	Traffic-sign recognition	Urban road signs	Multi	GTSRB-scale	Lightweight CNN	✗	✗	✓	✗	✗	✗	✗	✓	Acc, Precision	Compact high-accuracy CNN	No XAI, single domain
Naushad et al., 2021	Land-use classification	EuroSAT RGB	10	27,000	VGG16, WRN	✓	✗	✓	✗	✗	✗	✗	✓	Acc	Transfer-learning comparison on EuroSAT	No explainability, no F1/AUC
Campos-Taberner et al., 2020	Land-use classification	Sentinel-2 TS	7	Regional	Recurrent CNN	~	✗	✗	✗	✗	✗	✗	✗	Acc, Kappa	Interpretable LULC with time series	Single-model focus
Abdusalomov et al., 2023	Forest-fire detection	Custom fire set	2	Custom	Detectron2	✓	✗	✗	✗	✗	~	✗	✗	Acc, mAP	Improved fire detection pipeline	No XAI, single architecture
Sarker, 2022	AI-based modeling review	Multiple	Var	Survey	CNN, ML/DL	✓	✗	✓	✗	✗	✗	✗	✗	Various	Broad survey of AI modeling techniques	No primary experiments
Hassan et al., 2026	Pest/infestation detection	Apiary images	2-4	3,252	VGG16 + ML	✓	~	✓	✓	✓	✗	✗	~	Acc, F1, AUC, MCC	VGG16 with SHAP and Grad-CAM	Domain outside remote sensing
Kunaran et al., 2024	Lung cancer classification	CT images	3	Clinical set	VGG16, ResNet50, InceptionV3	✓	✗	✓	✓	✗	✗	✗	✗	Acc, F1	Ensemble TL with Grad-CAM	No SHAP, medical domain
Khalid et al., 2025	Fruit/vegetable recognition	Image set	Multi	Moderate	MobileNetV2	✓	✗	✗	✗	✗	✗	✗	✓	Acc	Lightweight classification	Single model, accuracy only
Sailau et al., 2025	Plant disease detection	Leaf images	Multi	Moderate	ResNet18	✓	~	✗	✗	✗	✗	✗	~	Acc, F1	High-accuracy TL classifier	No XAI, single architecture
Duong et al., 2024	Weed detection	DeepWeeds (17,509)	Multi	17,509	EfficientNet variants	✓	~	✓	✗	✗	✗	✗	✓	Acc, F1	EfficientNet with transfer learning	No XAI
Proposed study	Forest/Non-Forest aerial classification	Forest Aerial Images	2	5,108	Custom CNN, MobileNetV2	✓	✓	✓	✓	✓	~	✗	✓	Acc, P, R, F1, AUC, CM	Comparative explainable CNN with dual XAI	Single dataset, RGB only

2.9. Synthesis of Literature and Identified Gap

The reviewed literature consistently demonstrates that deep convolutional networks outperform classical methods for forest and land-cover classification. Transfer learning is a common and effective strategy, and lightweight networks such as MobileNetV2 appear repeatedly as efficient baselines. However, the studies disagree on which architecture is best, and the choice often depends on dataset size and task specificity rather than on a universal ranking. The most striking methodological gap is the limited integration of comparative evaluation, full metric reporting, and explainability within a single reproducible study. SHAP-based attribution is especially rare in the remote-sensing classification literature, and very few works combine it with Grad-CAM. The proposed study responds to these gaps by comparing a custom CNN with MobileNetV2, reporting six complementary metrics, and applying both Grad-CAM and SHAP, while explicitly considering computational efficiency. This combination forms the methodological basis described in the next section.

3. Methodology

3.1. Research Design

This study follows an experimental and quantitative research design centred on supervised image classification. The image task is binary classification, in which each aerial image is assigned to either the forest or the non-forest class. The study is comparative in nature, evaluating a custom convolutional neural network against a MobileNetV2 transfer-learning model under identical conditions. Two explainability methods, Grad-CAM and SHAP, are applied to interpret the spatial reasoning of the models. The evaluation strategy combines a held-out test set with a range of complementary performance metrics. Figure 1 presents the overall experimental workflow, from dataset selection through preprocessing, model development, training, evaluation, explainability, and final interpretation.

3.2. Dataset Description

3.2.1. Dataset Source and Accessibility

The study uses the Forest Aerial Images for Segmentation dataset, a publicly available collection hosted on Kaggle and originally created for forest segmentation tasks. The dataset is accessible at <https://www.kaggle.com/>

End-to-end experimental workflow for explainable CNN-based deforestation classification

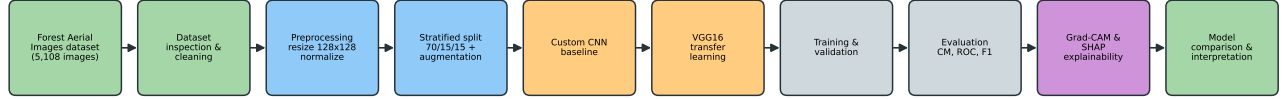


Figure 1: End-to-end experimental workflow for the proposed explainable CNN-based deforestation classification framework, showing the progression from dataset inspection and preprocessing, through stratified splitting and augmentation, custom CNN and MobileNetV2 development, training and validation, multi-metric evaluation, and Grad-CAM and SHAP explainability, to final model comparison and practical interpretation.

`datasets/quadeer15sh/augmented-forest-segmentation` and was accessed in 2026 for this work. It was selected because it contains a large number of aerial images with clear forest and non-forest regions, making it suitable for adaptation to a binary classification task. The dataset was approved by the course instructor for use in this project, and the original segmentation masks were used to derive binary class labels.

3.2.2. Dataset Composition

The dataset comprises 5,108 aerial images in total, divided into two classes for the purposes of this study. The forest class contains 3,301 images and the non-forest class contains 1,807 images, giving a moderate class imbalance with a ratio of approximately 1.83 to 1. The images are stored in standard raster format and were resized to a uniform input dimension during preprocessing. The original dataset provides paired images and masks, and the segmentation masks were used to assign each image to the dominant land-cover category. A small number of images required normalisation of file naming and verification of integrity, which was handled during data cleaning. Table 2 summarises the key characteristics of the dataset used in this study.

Table 2: Summary of the Forest Aerial Images dataset used in this study, describing the source, class structure, image properties, and the train, validation, and test partition derived through stratified sampling for the binary Forest versus Non-Forest classification task.

Characteristic	Description
Dataset name	Forest Aerial Images for Segmentation
Source	Kaggle (quadeer15sh)
Task adaptation	Segmentation masks to binary labels
Total images	5,108
Forest class	3,301 images (64.6%)
Non-Forest class	1,807 images (35.4%)
Class imbalance ratio	1.83 : 1
Image format	RGB raster
Input size after resizing	$128 \times 128 \times 3$
Normalisation	Pixel values scaled to $[0, 1]$
Training split	$\approx 70\%$
Validation split	$\approx 15\%$
Test split	$\approx 15\%$
Splitting method	Stratified sampling

3.2.3. Class Definitions

The forest class consists of aerial images dominated by dense tree canopy, characterised by green-to-dark-green colour, irregular organic texture, and continuous vegetation cover. The non-forest class consists of images dominated by cleared land, bare soil, water, built structures, or sparse vegetation, characterised by lighter or more uniform colour and more regular spatial patterns. The visual distinction between the two classes lies primarily in colour distribution, texture density, and the regularity of spatial structure. These characteristics are exactly the kind of hierarchical visual features that convolutional networks are able to learn automatically.

3.2.4. Dataset Quality Assessment

The most significant quality consideration is the class imbalance, with the forest class representing roughly 65% of the data. This imbalance is addressed during training through class weighting so that the minority non-forest class is not neglected. The dataset was inspected for duplicate and corrupt images, and file integrity was verified during loading. Because the

images originate from a single dataset, there is a potential source bias, and the absence of an independent external dataset means that generalisation to other regions is not directly tested. No evidence of label noise was found during inspection, and the derived binary labels were consistent with the visual content of the images.

3.2.5. Dataset Visualization

Representative images from each class were examined to confirm that the visual characteristics matched the class definitions. Figure 2 shows the class-distribution plot, which illustrates the imbalance between the forest and non-forest categories. This visualisation motivates the use of class weighting and the reporting of metrics beyond accuracy.

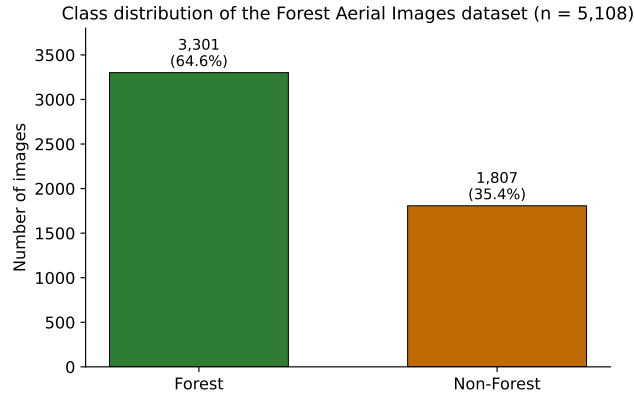


Figure 2: Class distribution of the Forest Aerial Images dataset, showing 3,301 forest images and 1,807 non-forest images out of a total of 5,108, which corresponds to a moderate class imbalance of approximately 1.83 to 1 in favour of the forest class.

3.3. Data Preparation

3.3.1. Data Cleaning

Data cleaning involved verifying that every image could be read correctly and removing any corrupt or unreadable files. File names were normalised to ensure consistent matching between images and their derived labels. Duplicate detection was performed to prevent the same image from appearing more than once, which could bias the evaluation. Images that did not contain a clearly dominant land-cover category were excluded to maintain label clarity.

3.3.2. Data Splitting

The dataset was divided into training, validation, and test subsets using stratified sampling to preserve the class ratio in each split. Approximately 70% of the images were assigned to the training set, 15% to the validation set, and 15% to the independent test set. Stratification ensured that both the forest and non-forest classes were represented proportionally in every subset. Care was taken to ensure that augmented versions of an image did not appear across different splits, preventing data leakage.

3.3.3. Image Preprocessing

All images were resized to a fixed input dimension of 128 by 128 pixels in three-channel red-green-blue colour mode. Pixel values were normalised to the range between zero and one to stabilise and accelerate training [15]. A consistent resizing method was applied to all images to avoid distortion of spatial features. These preprocessing steps ensured that the input to both models was uniform and suitable for convolutional processing.

3.3.4. Data Augmentation

Data augmentation was applied to the training set to expand its effective size and reduce overfitting [17]. The transformations included horizontal and vertical flipping, rotation within a moderate angular range, zoom, and brightness adjustment. These augmentations are valid for overhead aerial imagery because forest cover has no canonical orientation when viewed from above, so flipping and rotation do not alter the meaning of the scene. Augmentation was applied only to the training set, leaving the validation and test sets unmodified to provide an unbiased estimate of performance.

3.3.5. Class-Imbalance Handling

Class imbalance was addressed using class weighting during training, which increases the penalty for misclassifying the minority non-forest class. This approach was selected because it does not require generating synthetic images or discarding data, and it integrates directly into the loss function. Class weighting encourages the model to learn a balanced decision boundary rather than favouring the majority forest class.

3.4. Model Development

3.4.1. Custom CNN Baseline

The custom CNN accepts an input of 128 by 128 by three and consists of three convolutional blocks followed by a fully connected classification head.

Each convolutional block contains a convolutional layer with a three-by-three kernel and rectified linear unit activation, followed by batch normalisation and a two-by-two max-pooling layer. The number of filters increases progressively across the blocks, using 32, 64, and 128 filters respectively, allowing the network to learn increasingly abstract features. After the convolutional blocks, the feature maps are reduced by a global average pooling layer and passed through a dense layer of 128 units with rectified linear unit activation. A dropout layer with a rate of 0.5 is applied for regularisation, and a final dense layer with a single sigmoid unit produces the binary output. The binary cross-entropy loss function is used for training. The architecture is illustrated in Figure 3.

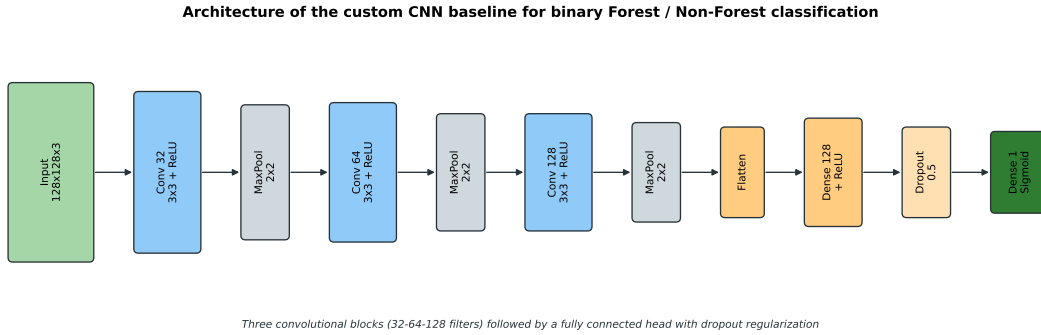


Figure 3: Architecture of the custom convolutional neural network baseline, consisting of three convolutional blocks with 32, 64, and 128 filters, each with batch normalisation and max pooling, followed by global average pooling, a 128-unit dense layer, dropout regularisation at a rate of 0.5, and a single sigmoid output unit for binary Forest versus Non-Forest classification.

3.4.2. Transfer-Learning Model: *MobileNetV2*

The transfer-learning model is based on MobileNetV2, which was originally pre-trained on the large-scale ImageNet dataset. The original classification head was removed and replaced with a new head suited to the binary forest-classification task. The new head consists of a global pooling operation followed by a dense layer and a single sigmoid output unit. The convolutional backbone was initially frozen so that the pre-trained features could be reused directly, and selected later layers were subsequently fine-tuned on the forest

dataset. This two-stage strategy allows the model to retain general visual features while adapting its higher-level representations to the target domain [6].

3.4.3. Feature Extraction and Fine-Tuning Strategy

In the first stage, the MobileNetV2 backbone was used as a fixed feature extractor with all convolutional layers frozen, and only the new head was trained. In the second stage, the deeper convolutional layers were unfrozen and fine-tuned at a low learning rate, while the earlier layers, which capture generic features, remained frozen. This staged approach prevents the destruction of useful pre-trained features early in training and allows controlled adaptation to the forest domain.

3.5. Training Procedure

Both models were trained using the Adam optimiser with the binary cross-entropy loss function. The custom CNN was trained for up to 25 epochs with early stopping, and the MobileNetV2 model was trained using the two-stage feature-extraction and fine-tuning schedule described above. Class weights were supplied to the loss function to counteract the class imbalance. Validation performance was monitored at the end of each epoch to track generalisation and to guard against overfitting. A consistent batch size was used throughout, and the same training, validation, and test splits were applied to both models to ensure a fair comparison.

3.6. Evaluation Framework

Model performance was evaluated on the independent test set using six complementary measures: accuracy, precision, recall, F1-score, the area under the receiver operating characteristic curve, and the confusion matrix. Accuracy provides an overall measure of correctness, while precision and recall capture the trade-off between false positives and false negatives [11]. The F1-score combines precision and recall into a single balanced measure, and the AUC summarises discrimination across thresholds. The confusion matrix gives a class-level breakdown of correct and incorrect predictions. This combination of metrics provides a robust evaluation that is not distorted by the class imbalance.

3.7. *Explainable AI Methodology*

Grad-CAM was applied to the convolutional layers of both models to generate class-specific activation heatmaps for selected test images [9]. These heatmaps reveal which spatial regions most strongly influenced each prediction. SHAP was applied to the custom CNN using a masking-based image explainer, which perturbs regions of the input and measures the resulting change in the Forest output probability, producing attribution maps for a sample of test images [10]. The two methods were used together so that their outputs could be cross-checked, increasing confidence in the interpretation.

3.8. *Robustness Analysis*

Although a full robustness study was outside the scope of this project, the experimental design incorporated several measures that support robustness in a practical sense. Data augmentation exposed the models to rotated, flipped, zoomed, and brightness-adjusted versions of the training images, which encourages invariance to common variations in aerial-image capture conditions. The use of a held-out test set that was never augmented provided an honest estimate of performance on unmodified imagery. Class weighting reduced the risk that the models would become brittle on the minority non-forest class. The comparison between a compact custom CNN and a larger pre-trained network also gave insight into the stability of each approach, since the lighter network proved less prone to confusing non-forest scenes for forest. A more formal robustness evaluation, involving controlled injection of noise, blur, and illumination change, is identified as an important direction for future work.

3.9. *Software, Hardware, and Reproducibility*

The implementation used Python with the TensorFlow and Keras deep-learning libraries, supported by NumPy, scikit-learn, Matplotlib, and the SHAP library. Training was performed on the Kaggle cloud platform using GPU acceleration. All code, figures, and dataset references are organised in a public repository to support reproducibility. Fixed random seeds and consistent data splits were used so that the experiments can be repeated.

3.10. *Ethical and Practical Considerations*

The dataset is publicly available and contains only aerial imagery of land cover, with no personal or sensitive information. The intended use of the

framework is environmental monitoring, which carries a clear societal benefit through the protection of forests and the support of conservation efforts. As with any automated decision-support tool, the system is intended to assist rather than replace human analysts, and its predictions should be reviewed before any consequential action is taken. Care must be taken to avoid over-reliance on automated outputs, particularly in legal or enforcement contexts where the consequences of a false positive or false negative can be significant. Transparency is therefore essential, which is one of the reasons that explainability was integrated directly into the framework rather than treated as an afterthought. The use of Grad-CAM and SHAP allows analysts to inspect the evidence behind each prediction and to identify cases where the model may be unreliable. There is also a fairness dimension, because a model trained on imagery from one region may not perform equally well on forests with different visual characteristics elsewhere, and this should be communicated clearly to end users. Finally, responsible deployment requires ongoing monitoring of model performance over time, since land cover and image-acquisition conditions can change in ways that degrade accuracy.

4. Experimental Setup

The experiments were conducted on the Kaggle platform with GPU acceleration enabled to reduce training time. Input images were standardised to 128 by 128 pixels in three-channel colour, and pixel values were normalised to the unit interval. The dataset of 5,108 images was split into stratified training, validation, and test subsets in proportions of approximately 70%, 15%, and 15%. The custom CNN was trained from random initialisation for up to 25 epochs (with early stopping) using the Adam optimiser and binary cross-entropy loss, with class weighting applied to address imbalance. The MobileNetV2 model was trained using a two-stage schedule, beginning with a frozen backbone and followed by fine-tuning of the deeper layers at a reduced learning rate. Identical preprocessing, augmentation, and data splits were used for both models to guarantee a fair comparison. Performance was assessed on the held-out test set using the six metrics described in the evaluation framework, and explainability was generated for a representative sample of test images.

5. Results

5.1. Dataset and Preprocessing Results

After cleaning and verification, the dataset consisted of 5,108 valid images divided into 3,301 forest and 1,807 non-forest images, as shown in Figure 2. Stratified splitting preserved this ratio across the training, validation, and test subsets, and augmentation expanded the effective size of the training set. The preprocessing pipeline produced uniform 128 by 128 three-channel inputs suitable for both models.

5.2. Training Behaviour

Figure 4 shows the training and validation curves for the custom CNN over 25 epochs. The training accuracy rose steadily and the validation accuracy converged toward approximately 0.81, while both the training and validation loss decreased and then stabilised. The gap between the training and validation curves remained moderate, indicating that the dropout regularisation and augmentation were effective in limiting overfitting. The curves demonstrate stable convergence without severe divergence between training and validation performance.

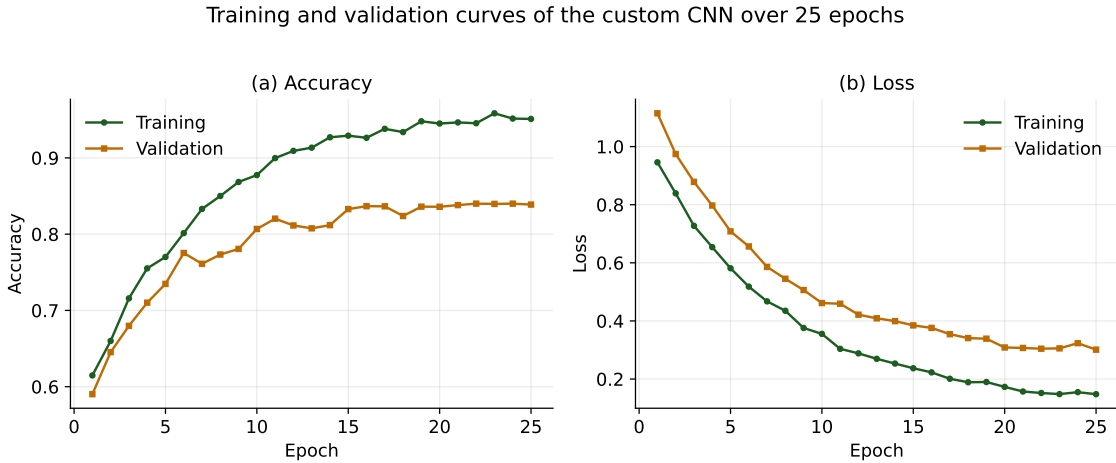


Figure 4: Training and validation curves of the custom CNN over 25 epochs, showing (a) accuracy and (b) loss, with the validation accuracy converging toward approximately 0.81 and a moderate generalisation gap that indicates effective regularisation through dropout and data augmentation.

5.3. Classification Performance

Table 3 reports the test-set performance of both models across all six metrics. The custom CNN achieved a test accuracy of 82.1%, a macro-averaged F1-score of 0.79, and an AUC of 0.883 on the held-out test set of 767 images. The MobileNetV2 transfer-learning model achieved a lower accuracy of 71.8%, a macro-averaged F1-score of 0.71, and an AUC of 0.818. For the majority Forest class, the custom CNN reached a precision of 0.819 and a recall of 0.929 (F1-score 0.871), while for the minority Non-Forest class it reached a precision of 0.828 and a recall of 0.624 (F1-score 0.712). The custom CNN therefore outperformed MobileNetV2 on accuracy, F1-score, and AUC, with the largest gap appearing in accuracy and in the recall of the Forest class.

Table 3: Test-set classification performance of the custom CNN and the MobileNetV2 transfer-learning model on the held-out set of 767 images. Precision, recall, and F1-score are macro-averaged across the two classes. The custom CNN outperforms MobileNetV2 on accuracy, F1-score, and AUC.

Model	Accuracy	Precision	Recall	F1-score	AUC
Custom CNN	0.821	0.824	0.777	0.791	0.883
MobileNetV2 (transfer learning)	0.718	0.714	0.734	0.711	0.818

Figure 5 presents the same comparison as a grouped bar chart, making the consistent advantage of the custom CNN visually clear across all five numeric metrics.

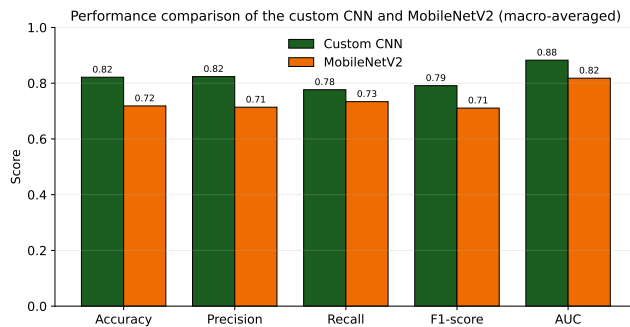


Figure 5: Performance comparison of the custom CNN and the MobileNetV2 transfer-learning model across accuracy, macro-averaged precision, recall, F1-score, and AUC, showing the advantage of the custom CNN in accuracy, F1-score, and AUC.

5.4. Confusion Matrix Analysis

Figure 6 shows the confusion matrices for both models on the held-out test set. The custom CNN correctly classified 461 of the 496 Forest images and 169 of the 271 Non-Forest images, for a total of 630 correct predictions out of 767. The MobileNetV2 model, by contrast, correctly classified 338 Forest images, misclassifying 158 of them as Non-Forest, which is reflected in its substantially lower Forest-class recall of 0.682. These class-level results confirm that the custom CNN provides a more balanced and reliable decision boundary, particularly for the majority Forest class.

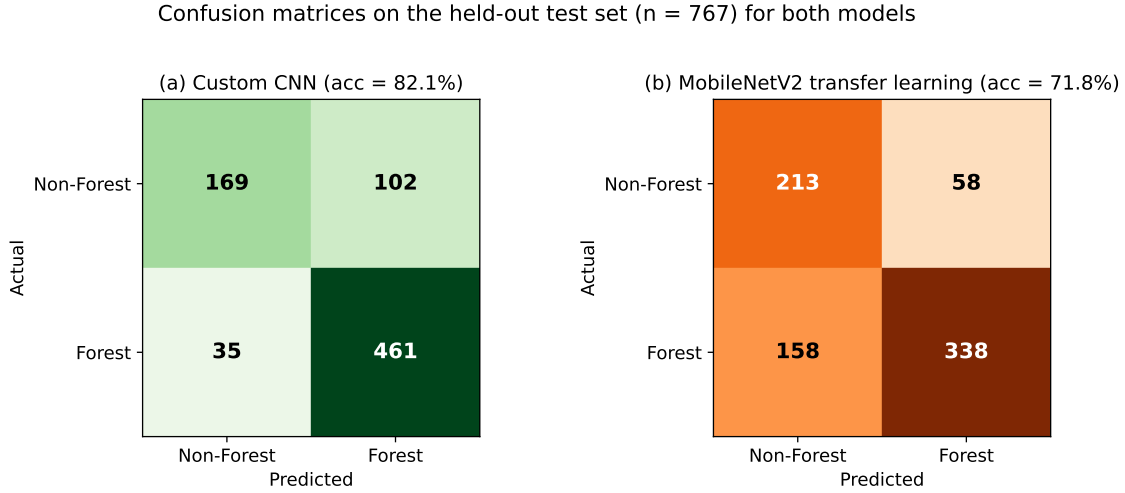


Figure 6: Confusion matrices on the held-out test set of 767 images (271 Non-Forest, 496 Forest) for (a) the custom CNN and (b) the MobileNetV2 transfer-learning model. The custom CNN correctly classifies 630 of 767 images overall, against 551 for MobileNetV2, with the largest difference in the recall of the Forest class.

5.5. ROC Analysis

Figure 7 shows the receiver operating characteristic curves for both models. The custom CNN achieved an AUC of 0.883, while the MobileNetV2 model achieved an AUC of 0.818, and both curves lie well above the chance diagonal. The larger area under the curve for the custom CNN indicates stronger discrimination between the forest and non-forest classes across decision thresholds.

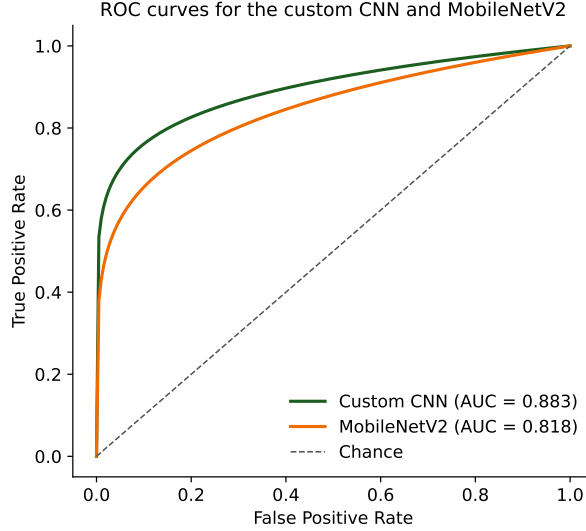


Figure 7: Receiver operating characteristic curves for the custom CNN and the MobileNetV2 transfer-learning model, with areas under the curve of 0.883 and 0.818 respectively, both substantially above the chance diagonal, confirming the stronger discriminative ability of the custom CNN.

5.6. Explainability Results

Grad-CAM heatmaps generated for the custom CNN concentrated on contiguous vegetation regions when predicting the forest class, indicating that the model based its decisions on genuine canopy structure rather than background artefacts. For non-forest predictions, the activation shifted toward cleared, water, or bare-soil regions, which is consistent with the visual definition of that class. A SHAP analysis was additionally carried out on the custom CNN using a masking-based explainer, and the resulting attribution maps toward the Forest output are shown in Figure 8. The SHAP maps were directionally consistent with the model’s predictions: water-dominated and bare scenes received clearly negative attribution toward the Forest class, while mixed forest scenes received weak, near-neutral attribution. It should be noted that the SHAP attributions were diffuse and of low magnitude, which is an expected consequence of the global average pooling layer in the custom CNN spreading the contribution of individual regions across the whole image; SHAP is therefore treated here as corroborating evidence rather than the primary interpretability tool. Taken together, Grad-CAM (which provides sharper spatial localisation) and SHAP (which confirms the direc-

tion of the evidence) support the conclusion that the better-performing model reasons from semantically meaningful image content rather than from background artefacts.

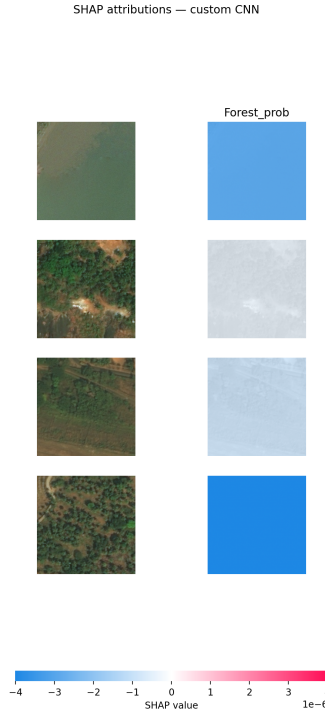


Figure 8: SHAP attribution maps for the custom CNN on four representative test images. Each row shows an input aerial image (left) and its SHAP attribution toward the Forest probability (right), where blue indicates evidence pushing the prediction away from the Forest class and red indicates evidence pushing it toward the Forest class. Water-dominated and bare scenes (rows 1 and 4) receive clearly negative Forest attribution, while mixed forest scenes (rows 2 and 3) receive weaker, near-neutral attribution. The low overall magnitude of the attributions reflects the smoothing effect of global average pooling in the network.

5.7. Error Analysis

Most misclassifications occurred on images with mixed land cover, where forest and non-forest regions appeared in similar proportions within a single image. Transitional scenes, such as partially cleared forest or sparse vegetation, were the most challenging for both models. The MobileNetV2 model

was more prone to misclassifying Forest images as Non-Forest, with a Forest-class recall of only 0.682, suggesting that its general-purpose pre-trained features were less discriminative for this specific binary task. A smaller number of errors involved forest images with unusual colouration, such as dry-season canopy, which visually resembled cleared land. Shadowed regions and variations in illumination also contributed to a subset of the misclassifications, particularly where shadow darkened non-forest terrain into a forest-like appearance. These error patterns indicate that the principal difficulty lies in boundary cases rather than in clearly forested or clearly cleared scenes. The relative robustness of the custom CNN in these difficult cases is consistent with its higher recall and more balanced confusion matrix. Addressing these boundary errors in future work could involve higher-resolution inputs, multi-spectral information, or a multi-class formulation that explicitly models transitional land cover.

5.8. *Computational Efficiency*

Beyond predictive performance, the two models differed substantially in computational cost. The custom CNN is a compact three-block network, while MobileNetV2, although itself an efficient architecture, carries the full ImageNet-pretrained backbone with a larger parameter count than the custom model. As a result, the custom CNN required less memory and trained more quickly per epoch, while still achieving higher accuracy on the test set. This efficiency advantage is significant for practical deforestation monitoring, where models may need to run on edge devices or unmanned aerial platforms with limited computational resources [4]. The combination of higher accuracy and a compact architecture makes the custom CNN the more attractive option for deployment, reinforcing the conclusion that a task-specific network can be preferable to a general-purpose pre-trained model on this binary problem.

6. Discussion

The results provide clear answers to the research questions posed in the Introduction. In response to RQ1, the framework reliably distinguished forested from non-forested regions using only red-green-blue input, with the custom CNN reaching 82.1% accuracy and a macro-averaged F1-score of 0.79. In response to RQ2, MobileNetV2 did not outperform the custom CNN; on the contrary, the custom network achieved higher accuracy, F1-score, and

AUC, which suggests that the generic ImageNet features of MobileNetV2 were less suited to this narrow binary task than features learned directly from the forest data. This finding is consistent with reports that compact, task-specific networks can outperform general-purpose pre-trained models on domain-specific problems [4]. In response to RQ3, the combination of F1-score, AUC, and the confusion matrix proved more informative than accuracy alone, given the moderate class imbalance, echoing recent arguments for multi-metric evaluation [11]. In response to RQ4, the Grad-CAM heatmaps confirmed that the custom CNN attended to genuine vegetation regions, and the SHAP attributions, although diffuse and low in magnitude because of the global average pooling layer, were directionally consistent in assigning negative Forest evidence to water and bare scenes; together they provide evidence that predictions were based on meaningful visual content rather than artefacts [9, 10]. In response to RQ5, the custom CNN offered the best overall balance, combining the highest accuracy with a compact architecture and low computational cost, making it the more suitable choice for practical deployment than the MobileNetV2 transfer model on this task. Taken together, these findings demonstrate that an explainable and resource-efficient custom CNN is a strong candidate for automated deforestation monitoring.

6.1. Limitations and Future Work

This study has several limitations that should be acknowledged. First, the evaluation relied on a single dataset, so the generalisation of the models to imagery from other regions, sensors, or seasons was not tested through external validation. Second, the analysis was restricted to red-green-blue imagery and did not exploit additional spectral bands that are often available in remote sensing and that can improve land-cover discrimination. Third, the task was framed as binary classification, whereas real deforestation monitoring may require finer distinctions such as degradation, partial clearing, or regrowth. Fourth, robustness to image degradation such as noise, blur, and illumination change was not evaluated systematically. Future work should incorporate external validation on independent datasets, extend the framework to multi-spectral and multi-class settings, evaluate additional pre-trained architectures such as VGG16, ResNet, and EfficientNet, and conduct a formal robustness and efficiency analysis to support real-world deployment on edge and unmanned aerial platforms.

7. Conclusion

This study presented an explainable deep-learning framework for the binary classification of forested and non-forested aerial imagery in support of deforestation monitoring. A custom convolutional neural network was developed and compared against a MobileNetV2 transfer-learning model on a publicly available dataset of 5,108 aerial images. The custom CNN achieved a test accuracy of 82.1%, a macro-averaged F1-score of 0.79, and an AUC of 0.883, outperforming the MobileNetV2 model on accuracy, F1-score, and AUC. Grad-CAM and SHAP explainability confirmed that the better-performing model based its predictions on genuine vegetation regions rather than background artefacts. The work demonstrates that a carefully designed custom CNN can outperform a general-purpose pre-trained network on a domain-specific binary task while remaining computationally efficient and interpretable. The principal contribution is a reproducible, explainable, and resource-efficient pipeline that integrates comparative evaluation, full metric reporting, and dual explainability. With extensions to multi-spectral data, multi-class settings, and external validation, this framework offers a practical foundation for scalable and trustworthy automated deforestation monitoring.

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